

GONZALO G. FERNANDEZ

Mechatronics Engineer | Firmware/Software Engineer

Fields of interest: Embedded systems, robotics and control theory

Programming languages: C, Python, Bash, Verilog

Technologies: Linux, AVR, STM32, FreeRTOS

Tools: Git, Matlab, KiCAD, STM32CubeMX, Vivado, SolidWorks

Languages: Spanish, English

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Mechatronics and software engineer interested in the design and development of software and firmware for embedded systems and robotics. Looking for an opportunity to start a career in the robotics industry.

Experience

Firmware QA Software Engineer – Marvell Technology – Córdoba, Argentina Oct 2022 – Now

- Software development for a firmware QA automation platform. Understanding of complex systems for its testing.
- Recruitment interviews and technical training of new team members. Communication with firmware developers.

Engineering internship – Simen Rayos X – Mendoza, Argentina Jan – Oct 2020

- Analysis, design, modification, assembly and documentation of PCBs for X-ray equipment.
- Use of KiCAD software. Programming of PIC and AVR microcontrollers (8/16 bit). I2C protocol analysis.

Teaching assistant – Universidad Nacional de Cuyo – Mendoza, Argentina Jan – Jul 2018

- Part-time position in the Mathematical Analysis II course.
- Prepared a joint assignment with the Physics II course and lecture notes on the Riemann–Stieltjes integral.

Education

Master degree in Embedded Systems – Universidad de Buenos Aires – Remote Feb 2023 – Now

- Currently in the Specialization in Embedded Systems, first year of the Master degree.
- Working in the design of a differential mobile robot prototype as platform for SLAM with ROS 2 through micro-ROS.

Mechatronics engineering – Universidad Nacional de Cuyo – Mendoza, Argentina Feb 2015 – Aug 2022

- Graduated with GPA of 8.83 out of 10.
- Final project: end-to-end development of a dual-SCARA parallel robotic manipulator.

Semester abroad – Universidad El Bosque – Bogotá, Colombia Feb – Jun 2022

- International study exchange scholarship in the Systems Engineering program.
- Courses on Compilers, Intelligent Systems, Microprocessors and Assembly, and Operating Systems.

Technician in Electromechanical Equipment and Installations – Esc. N° 4-228 "Ing. E. Izsaky" 2009 – 2014

- Graduated with GPA of 9.29 out of 10.
- Technical secondary school, oriented to the production of goods and services.

Research & Projects

Development of a dual-SCARA parallel robot – Universidad Nacional de Cuyo Feb 2019 – Aug 2022

- Type C research project of the Universidad Nacional de Cuyo. Mechatronics Engineering degree thesis.
- Design, manufacturing, simulation, and control of dual-SCARA parallel robot. Mainly involved in the mechanical design.

Sun tracking system for parabolic solar collector – Universidad Nacional de Cuyo Jul 2019 – Aug 2022

- Collaboration in PhD on the design and construction of a sun tracking system for a parabolic solar collector.
- Embedded system design, previous software integration on Raspberry Pi and mechanical prototype design.

Optimal LQG controller design for self-balancing robot – Universidad Nacional de Cuyo Dec 2020 – Oct 2021

- Project of the subject Control Systems. Poster in the XIX Working Meeting on Information Processing and Control.
- Self-balancing robot analysis and LQG optimal controller design with MATLAB Simulink and Simscape Multibody.

FreeRTOS in EDU-CIAA for robotic arm control – Universidad Nacional de Cuyo Oct 2020 – May 2021

- Robotic arm control using FreeRTOS on the EDU-CIAA development board (Argentine open-source platform).
- NXP LPC4337 microcontroller: dual core ARM Cortex-M4F and Cortex-M0.

Courses

Advanced Digital Design – Fulgor Foundation – Córdoba, Argentina Aug – Dec 2022

- Organized by the Fulgor Foundation and taught by Dr. Ariel L. Pola, 75 hours long.
- Advanced concepts of digital design in FPGA and VLSI for application in DSP systems.

Basic Digital Design – Fulgor Foundation Mar 2021

- Organized by the Fulgor Foundation and taught by Dr. Ariel L. Pola in the context of EAMTA 2021.